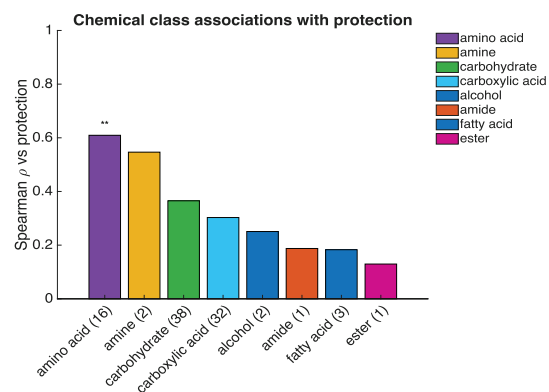
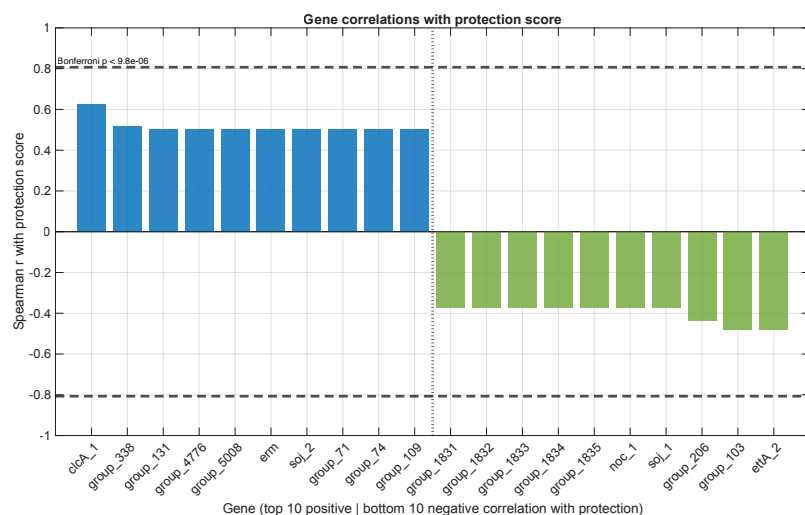




Supplementary Figure 4A: Spearman correlation between individual gene presence/absence and protection score across the 21 ST1 clinical isolates. The top 10 positively (blue) and bottom 10 negatively (green) correlated genes are shown. Dashed horizontal lines indicate the Bonferroni-corrected significance threshold ($p < 9.8 \times 10^{-6}$). The dotted vertical line separates positive from negative correlations. Gene identifiers are shown on the x-axis. (B) Mean Spearman correlation between protection score and carbon source utilization, grouped by chemical class from Biolog PM1 data. The number of carbon sources within each class is indicated in parentheses on the x-axis. Asterisks denote statistical significance ($p < 0.05$). Chemical classes are color-coded as indicated in the legend.